

Fake News Detection: A Comparative Study of TF-IDF, DistilBERT and BERT with Frozen and Fine-tuned Strategies

Python for NLP — Final Project Report

Seifeddine Ben Yaala · University ID: 3511613
Heinrich-Heine-Universität Düsseldorf

1. Introduction

The rapid growth of digital media has transformed how information is produced and consumed. Social networks, news aggregators and messaging platforms now distribute millions of articles every day, most of them faster than any human editor could verify. Alongside legitimate reporting, this ecosystem has enabled the widespread circulation of fabricated or deliberately misleading content — commonly referred to as *fake news* — with documented consequences on elections, public health, and trust in institutions (Shu et al., 2017).

Manual fact-checking cannot keep up with this volume, which motivates the development of automatic detection systems. Machine learning classifiers trained to distinguish fabricated articles from legitimate ones can flag suspicious content at scale, prioritise items for human review, or serve as a first filter in content-moderation pipelines. The quality of such systems — both in accuracy and in operational cost — directly determines whether they are practical to deploy.

Approach and research questions

Modern NLP offers a broad spectrum of options for text classification, from simple bag-of-words classifiers such as TF-IDF combined with logistic regression, to large pretrained Transformer models such as BERT and its lighter variant DistilBERT. Each option comes with a different trade-off between predictive performance and computational cost, and it is not obvious in advance which one is worth deploying for fake news detection.

This project compares five models on the same dataset under identical conditions: a TF-IDF + Logistic Regression baseline, DistilBERT and BERT-base each in a frozen configuration (only the classification head is trained) and a fine-tuned configuration (all parameters are updated). Predictive metrics (accuracy, macro-F1) are reported alongside computational cost (training time, inference time, GPU memory, parameter count), so the accuracy-cost trade-off can be discussed quantitatively.

The main research questions are:

- **Q1:** Does fine-tuning outperform freezing the pretrained encoder for fake news detection?
- **Q2:** Does the larger BERT model justify its higher computational cost compared to the smaller DistilBERT?

2. Background and Related Work

The three families of approaches compared in this study are presented below.

- **TF-IDF (Term Frequency – Inverse Document Frequency):** represents each document as a sparse weighted bag-of-words vector, where each term is weighted by how frequently it appears in the document relative to the corpus. Combined with a linear classifier such as logistic regression, TF-IDF is a strong baseline for text classification tasks where topical vocabulary is informative.
- **BERT (Devlin et al., 2019):** a Transformer encoder pretrained on masked language modelling and next-sentence prediction. In this study, BERT-base has 109.5M parameters across 12 Transformer layers.
- **DistilBERT (Sanh et al., 2019):** a compressed version of BERT-base obtained through knowledge distillation, retaining approximately 97% of BERT’s performance on GLUE while using ~40% fewer parameters and running ~60% faster. It has 67.0M parameters across 6 layers.

Pretrained encoders can be used in two main modes (Howard and Ruder, 2018). In the **frozen** configuration, encoder weights are held fixed and only a task-specific classification head on top of the [CLS] representation is trained. In the **fine-tuned** configuration, all encoder parameters are also updated. Fine-tuning typically achieves higher accuracy at the cost of more time, memory and overfitting risk.

3. Data

The dataset is *GonzaloA/fake_news* (GonzaloA, 2021), available on the HuggingFace Datasets hub under the CC BY 4.0 licence. It contains English-language news articles labelled as FAKE (0) or REAL (1). Preprocessing consisted of the following steps:

- lowercasing;
- removal of URLs and email addresses;
- removal of non-alphabetical characters;
- whitespace collapse;
- deduplication and dropping of empty articles.

To reduce computational cost, the training set was randomly subsampled to 10,000 articles using a fixed seed.

Splits and class distribution

The 10,000 subsampled training articles were further split into 9,000 training and 1,000 validation using a stratified split (seed = 42). The provided test set (8,101 articles after cleaning) was held out and used only for final evaluation. The training subset contained 5,467 REAL and 4,533 FAKE articles — reasonably balanced — so accuracy is a meaningful primary metric, with macro-F1 reported as a per-class check. Figure 1 summarises the class distribution and the article-length distribution in the training set.

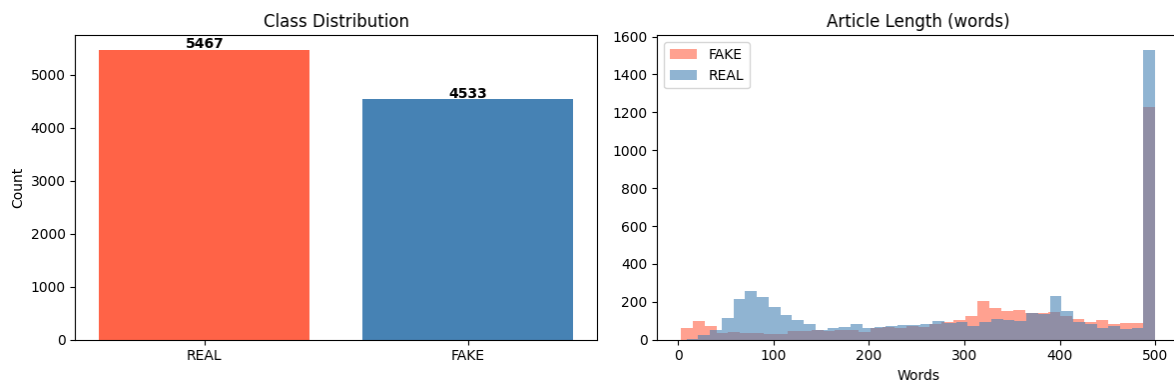


Figure 1: Left: class distribution in the training set. Right: distribution of article lengths in words, split by class.

Figure 1 reveals two notable patterns in the training data:

- **Class distribution (left panel):** shows a mild imbalance of roughly 55% REAL to 45% FAKE. This ratio is close enough to 50/50 that accuracy is a fair metric on its own, but it also means that a model which collapses to predicting REAL for almost every article can still reach approximately 55% accuracy without learning anything useful about the task. The macro-F1 metric penalises this failure mode explicitly.
- **Article-length distribution (right panel):** shows a large spike at the 500-word cap for both classes, meaning many articles are longer than displayed. REAL articles (blue) are bimodal — concentrated at short briefs under 150 words and long reports near the cap — while FAKE articles (red) show a broader spread between 100 and 400 words. On average, FAKE articles are slightly longer (424 words) than REAL articles (386 words). The practical implication is that truncating inputs to 256 tokens for the Transformer models drops informative content from many articles.

Limitations of the dataset

Three main limitations should be kept in mind when interpreting the results:

- the dataset consists mainly of English political articles, so results may not transfer to other domains or languages;
- articles were truncated to 256 tokens for the Transformer models, which may drop informative content;
- labels are assumed correct but not independently verified.

4. Methodology and Experimental Setup

4.1 Models

Five models were compared under identical experimental conditions. The Transformer models were implemented using the HuggingFace Transformers library (Wolf et al., 2020):

- **TF-IDF + Logistic Regression (baseline):** scikit-learn (Pedregosa et al., 2011) TfidfVectorizer with `max_features = 50,000`, `ngram_range = (1, 2)`, `sublinear_tf = True`; LogisticRegression with `C = 1.0`, `max_iter = 1,000`.
- **DistilBERT Frozen:** distilbert-base-uncased with encoder frozen; only the pre_classifier and classifier heads (592,130 parameters) trained.
- **DistilBERT Fine-tuned:** distilbert-base-uncased with all 66,955,010 parameters updated.
- **BERT Frozen:** bert-base-uncased with encoder frozen; only the classifier head (1,538 parameters) trained.
- **BERT Fine-tuned:** bert-base-uncased with all 109,483,778 parameters updated.

4.2 Hyperparameters and evaluation

To isolate the effect of the studied variables, all Transformer models were trained under identical conditions:

- maximum sequence length = 256 tokens, batch size = 16;
- learning rate = $2e-5$ with AdamW (`weight_decay = 0.01`);
- linear warmup for 10% of steps followed by linear decay;
- gradient clipping at 1.0;
- 3 training epochs with early-stopping (patience 2) on validation macro-F1;
- a single random seed (42) fixed for Python, NumPy and PyTorch (Paszke et al., 2019);
- all experiments executed on a Google Colab T4 GPU (16 GB VRAM).

Predictive performance is reported using accuracy and macro-F1 on the test set. Computational cost is measured through training time (wall-clock), inference time on the full test set, peak GPU memory, and parameter count. The only differences between conditions were the encoder architecture (BERT vs DistilBERT) and the adaptation strategy (frozen vs fine-tuned) — all other hyperparameters, splits, tokenizer settings and preprocessing steps were kept identical.

5. Results

Table 1 presents the final test-set performance and computational cost for all five models. It is worth noting that validation macro-F1 scores stayed within ± 0.5 points of the test scores throughout, which suggests the model selection process did not overfit to the validation set.

Table 1: Test-set performance and computational cost across the five models.

Model	Accuracy	F1 Macro	Train (s)	Inference (s)	GPU (GB)	Params (M)
TF-IDF + LR	0.9680	0.9679	20.9	0.008	0.00	0.0
DistilBERT Frozen	0.9124	0.9117	266.6	72.0	0.84	67.0
DistilBERT Fine-tuned	0.9863	0.9862	711.4	69.0	2.76	67.0
BERT Frozen	0.6803	0.6321	495.2	131.5	2.19	109.5
BERT Fine-tuned	0.9846	0.9845	1383.8	131.3	5.75	109.5

5.1 Overall performance comparison

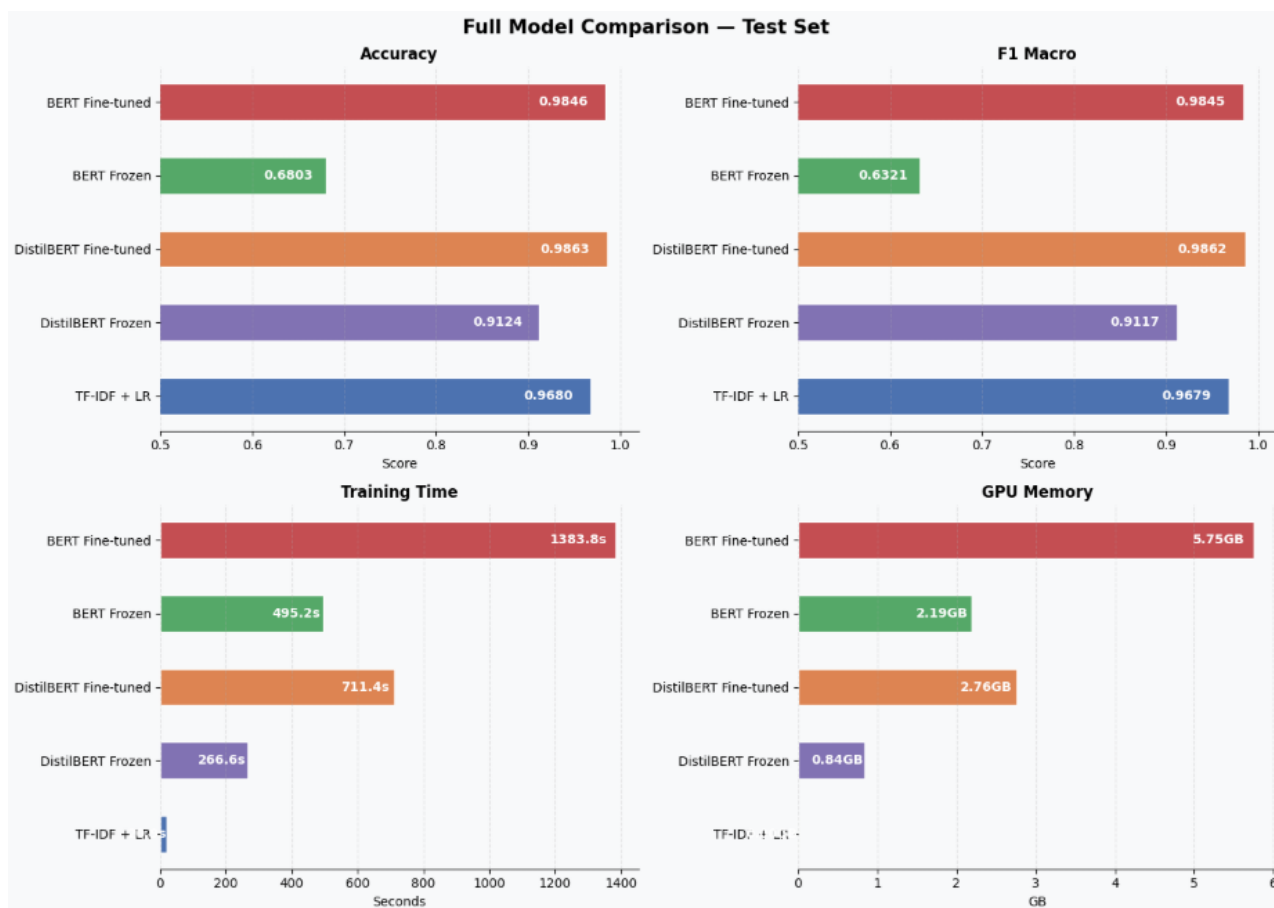


Figure 2: Full model comparison on the test set

Looking at Figure 2, a fairly clear tier structure emerges among the five models. The two fine-tuned Transformers sit at the top, both around 98.5%, with TF-IDF close behind at 96.8%. DistilBERT Frozen lands in a middle tier at 91.2%, while BERT Frozen trails far behind at just under 68%. What stands out is that accuracy and F1 track each other closely for four of the five models — meaning none of them is quietly sacrificing one class to inflate the other. BERT Frozen breaks this pattern: its F1 score (0.63) drops well below its accuracy (0.68), a red flag for asymmetric behaviour that becomes obvious once we look at its confusion matrix later in this section.

The bottom row of the figure tells a different but equally important story: cost. BERT Fine-tuned needs 1,384 seconds to train — about 66 times longer than TF-IDF’s 21 seconds, and nearly double DistilBERT Fine-tuned’s 711 seconds. The same pattern shows up in memory usage, where BERT Fine-tuned peaks at 5.75 GB compared to 2.76 GB for its DistilBERT counterpart. Put together, these two rows point to fine-tuned DistilBERT as the most balanced option overall: nothing beats it on accuracy without costing noticeably more, and nothing is cheaper without giving up meaningful accuracy. Perhaps the most interesting takeaway from this tier structure is that *how* a model is adapted (frozen vs fine-tuned) matters more than *which* architecture is used (BERT vs DistilBERT) — the gap between the two frozen models is far wider than the gap between the two fine-tuned ones.

5.2 Per-model confusion matrix analysis

To understand where each model succeeds and fails at the class level, this section examines the confusion matrix of each of the five models in turn.

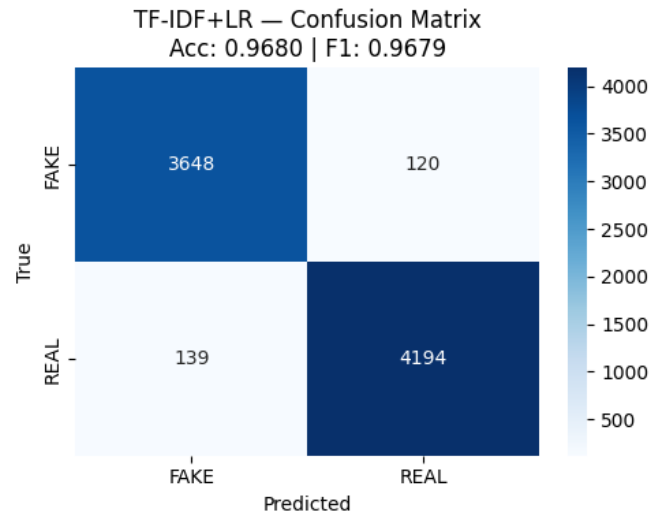


Figure 3: *TF-IDF + LR — confusion matrix on the test set.*

- TF-IDF + LR correctly classifies 3,648 of 3,768 FAKE articles (96.8% recall) and 4,194 of 4,333 REAL articles (96.8% recall).
- Errors are almost perfectly symmetric — 120 FAKE-as-REAL and 139 REAL-as-FAKE — showing the model is not biased toward either class.
- Given how simple this model is (a sparse vocabulary lookup and a single linear decision), this symmetry indicates fake and real articles in this dataset differ substantially in vocabulary, and these differences are stable across many articles rather than concentrated in a few unusual words.

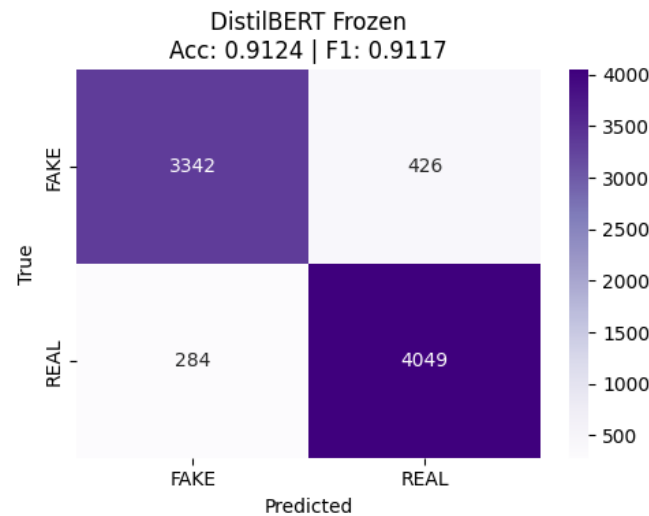


Figure 4: *DistilBERT Frozen — confusion matrix on the test set.*

- Freezing DistilBERT reduces accuracy to 91.2% and produces asymmetric errors: 426 FAKE articles are missed (labelled REAL), while only 284 REAL articles are labelled FAKE.
- FAKE recall drops to 88.7%, noticeably below the REAL recall of 93.4%.
- Without adapting the encoder, the mean [CLS] vector of a fake article is not clearly separated from that of a real one, and the classifier defaults toward the majority pattern.
- Interestingly, this frozen model still beats TF-IDF on some hard examples but misses many easier ones — a common trade-off when a fixed representation is imperfectly matched to the task.

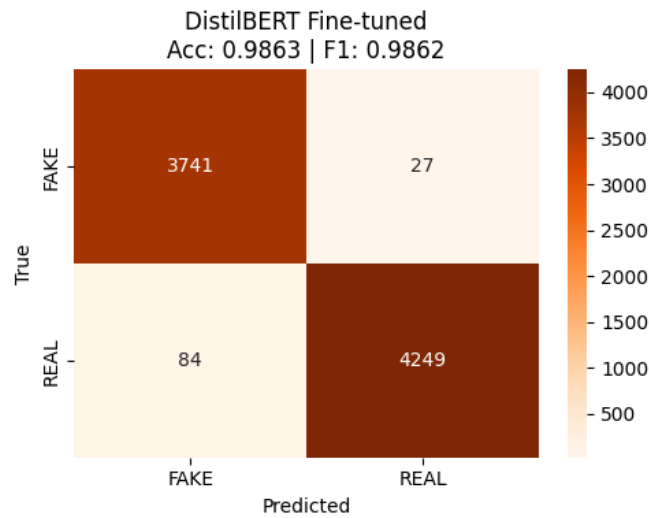


Figure 5: *DistilBERT Fine-tuned* — confusion matrix on the test set.

- Fine-tuning transforms DistilBERT’s behaviour: the model reaches 98.6% accuracy with a near-perfect FAKE recall of 99.3% (only 27 FAKE misclassified as REAL).
- Residual errors are concentrated on the REAL class, where 84 articles are labelled as FAKE.
- The very low false-negative rate on FAKE is desirable in a fake-news context, where missing fabrication is often worse than falsely flagging a real article.
- The asymmetry has flipped compared with the frozen version — the fine-tuned model is now over-suspicious rather than over-permissive — consistent with having learned strong lexical and structural cues for fabrication and applying them slightly too aggressively on polemical real articles.

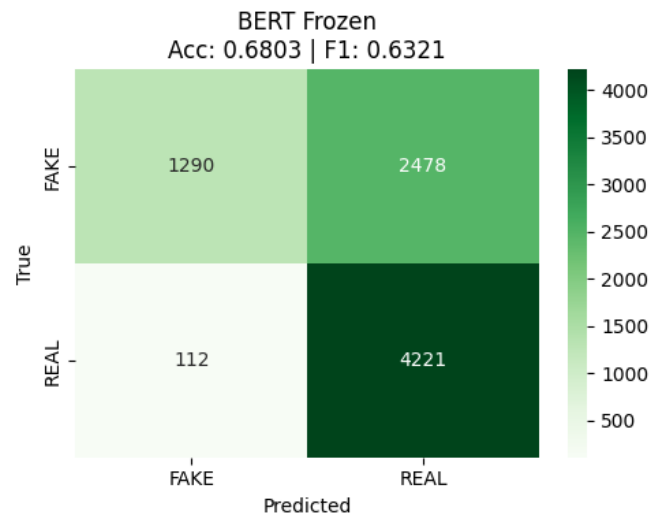


Figure 6: *BERT Frozen* — confusion matrix on the test set.

- This is the most revealing matrix. BERT Frozen classifies only 1,290 of 3,768 FAKE articles correctly — a FAKE recall of just 34.2% — while correctly labelling 4,221 of 4,333 REAL articles (97.4% recall).
- The model is essentially predicting REAL for almost every article: 6,699 of its 8,101 predictions (82.7%) are REAL.
- This is the direct explanation for its poor F1 (0.63): accuracy is inflated by the slight class imbalance, but performance on FAKE is barely better than random.
- The reason is technical: only the classifier layer (1,538 parameters — a single linear layer) is trainable, sitting directly on top of a pretrained [CLS] vector optimised for next-sentence prediction, not fake-news detection.
- With so few free parameters and a static input representation, the model has almost no capacity to learn a useful boundary and collapses to a majority-class predictor. The training set skew toward REAL (55/45) plausibly encourages this collapse.

- In contrast, DistilBERT’s classification head includes an additional `pre_classifier` layer that gives Frozen DistilBERT a fighting chance.

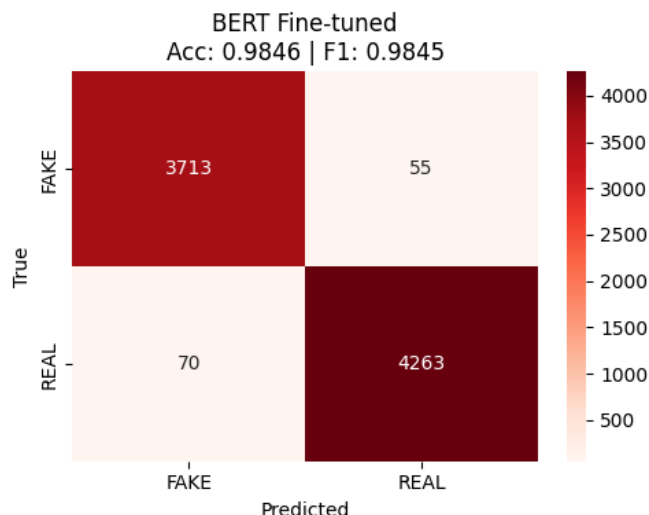


Figure 7: BERT Fine-tuned — confusion matrix on the test set.

- With fine-tuning enabled, BERT recovers dramatically: 3,713 FAKE (98.5% recall) and 4,263 REAL (98.4% recall) are correctly classified.
- The two error types are nearly balanced (55 FAKE-as-REAL, 70 REAL-as-FAKE), showing BERT’s larger capacity is used symmetrically.
- Compared with DistilBERT Fine-tuned, BERT is slightly less aggressive on FAKE (55 vs 27 misses) but slightly more accurate on REAL (70 vs 84 false alarms) — the two models are making genuinely different trade-offs even though their overall accuracy is close.
- In a deployment where false-negative fakes matter most, DistilBERT is preferable; where false-positive real articles matter most, BERT is marginally better. Neither difference is large enough to change the overall recommendation.

6. Discussion, Error Analysis and Limitations

6.1 Does fine-tuning outperform freezing?

Yes, clearly. Fine-tuning improved DistilBERT accuracy by 7.4 points and BERT accuracy by 30.4 points. The gap is much larger for BERT because BERT Frozen only trains 1,538 parameters, whereas DistilBERT Frozen has a `pre_classifier` layer that adds ~592k trainable parameters. As shown in Section 5.2, BERT Frozen effectively collapses to a majority-class predictor, while DistilBERT Frozen retains meaningful discrimination. Once the encoder is allowed to adapt, both models converge to essentially the same high accuracy (~98.5%).

6.2 Does BERT justify its cost over DistilBERT?

No. Fine-tuned DistilBERT (98.63%) very slightly outperformed fine-tuned BERT (98.46%), while using roughly half the parameters (67.0M vs 109.5M), less than half the training time (711 vs 1,384 seconds) and less than half the GPU memory (2.76 GB vs 5.75 GB). On this dataset, DistilBERT is Pareto-dominant: it matches or exceeds BERT on accuracy at a fraction of the cost.

6.3 The strength of the TF-IDF baseline

A striking observation is that TF-IDF + LR reaches 96.80% accuracy — only 1.8 points below the best Transformer. This suggests fake news articles in this dataset can be identified largely by their vocabulary: certain words, phrases and stylistic markers appear disproportionately often in fabricated content, and TF-IDF is highly effective at capturing them. The baseline trains in 21 seconds and infers in 8 milliseconds, versus 12 minutes of training and over one minute of inference for the best Transformer. In a production setting where inference cost matters, TF-IDF may be the practical choice.

6.4 Error analysis across the three main models

Of the 8,101 test articles, 7,778 were correctly classified by all three main models (TF-IDF, DistilBERT Fine-tuned, BERT Fine-tuned), 163 were misclassified only by TF-IDF, and 62 were misclassified by all three. In the first category — where both Transformers succeed but TF-IDF fails — articles typically use

vocabulary that overlaps with legitimate reporting (references to CIA, FBI, elections, senators) but appear in a fabricated context that requires understanding sentence structure, which Transformers capture but TF-IDF cannot.

The 62 hardest examples are more revealing. Manual inspection shows they are mostly REAL articles labelled as FAKE by every model. Many are politically polemical opinion pieces or highly editorialised reporting whose stylistic markers resemble those of fabricated content in the training set. This reflects a genuine ambiguity in the task: the boundary between polemical journalism and fabrication is not always clear, and models may inherit the ambiguity present in the dataset labels themselves.

6.5 Limitations

- Only one random seed was used. Ideally each experiment would be repeated with multiple seeds to report mean $\pm$ standard deviation.
- The training set was subsampled to 10,000 articles. The relative ranking may change with more data.
- Articles were truncated to 256 tokens, which may drop informative content at the end of long articles.
- The dataset is limited to English political news; conclusions may not transfer to other domains or languages.
- Label quality was not independently verified; ambiguous labels may bound the achievable accuracy.

7. Conclusion

This project compared five models for fake news detection on the GonzaloA/fake_news dataset under identical training conditions. Two conclusions stand out. First, fine-tuning is essential for these Transformer encoders on this task: freezing BERT reduces accuracy by 30 percentage points and produces a near-majority-class predictor, while freezing DistilBERT loses 7 points but still preserves useful discrimination. Second, the larger BERT does not justify its cost over DistilBERT: fine-tuned DistilBERT slightly exceeded BERT's accuracy while using half the memory and half the training time. A third and unexpected observation is that a plain TF-IDF + LR baseline reaches within 1.8 points of the best Transformer, at roughly a hundred-thousandth of the training cost — a reminder that model choice should follow from the accuracy-cost trade-off, not from architectural fashion. Realistic future work includes running each experiment with multiple seeds, sweeping the training-data size to build learning curves, comparing calibrated confidence scores, and studying how much of the TF-IDF baseline's strength is driven by dataset-specific stylistic cues.

Declaration of AI Usage

I used Claude (Anthropic) in three ways: to brainstorm the project design, to draft initial versions of the code, and to help structure and phrase this report. All code was reviewed and adapted by me until it ran end-to-end in Colab. Every number and figure in this report comes from that run, and I can explain every methodological decision. No results, references or datasets were fabricated with AI assistance, and AI tools are not cited as sources for any scientific claim.

References

- Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. (2019). *BERT: Pre-training of deep bidirectional transformers for language understanding*
- GonzaloA (2021). *Fake news dataset*
- Howard, J., and Ruder, S. (2018). *Universal language model fine-tuning for text classification*
- Paszke, A. et al. (2019). *PyTorch: An imperative style, high-performance deep learning library*.
- Pedregosa, F. et al. (2011). *Scikit-learn: Machine learning in Python*.
- Sanh, V., Debut, L., Chaumond, J., and Wolf, T. (2019). *DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter*
- Shu, K., Sliva, A., Wang, S., Tang, J., and Liu, H. (2017). *Fake news detection on social media: A data mining perspective*
- Wolf, T. et al. (2020). *Transformers: State-of-the-art natural language processing*